

LLMs for Survey Pretesting: How Well Can Large Language Models Identify Flaws in Survey Questions?

Abstract

This paper evaluates large-language-model (LLM) question-review based on simulated cognitive interviewing and expert review. We assess three prompt variants for each approach using a set of 20 questions containing deliberately embedded problems, with 7 items from the European Social Survey used as a control set. The best-performing method, guided cognitive testing, detects 75% of deliberately embedded problems and across three repeated runs it averages 70%, with estimates ranging from 65% to 79%. As with their human comparators, the cognitive-testing and expert-review methods tend to detect different kinds of problems, although these contrasts cannot be attributed uniquely to the human procedures on which they are modelled because the configurations also differ in prompt structure and processing. Detection and false-positive rates are sensitive to prompt design and LLM used: a structured prompt that works well on open-weight LLMs perform poorly on GPT-4o, and open-weight LLMs generally achieve higher detection rates but much poorer discrimination between flawed and well-designed questions. A full evaluation of a 20-item questionnaire completes in under an hour at a cost of a few dollars. These findings suggest that LLM-based question testing has the potential to provide a useful complement to human pretesting, particularly for early-stage screening of large item sets and in situations where resource constraints mean human testing is not possible.

1 Introduction

Cognitive interviewing has become the pre-eminent tool for evaluating draft survey questions before fielding a survey. Its distinctive value lies in anchoring diagnosis in the respondent’s perspective: participants think aloud while answering a question, or respond to targeted probes about their internal thought processes, and a trained analyst identifies difficulties across interviews. Drawing on the four-stage cognitive model of survey response developed by Tourangeau, Rips and Rasinski (2000), the method is designed to reveal how respondents actually navigate each cognitive stage, surfacing problems that may not be apparent from the question text alone (Willis 2005; Beatty and Willis 2007). Over the past several decades, it has moved from a relatively specialised technique to a routine component of questionnaire development, with its use growing substantially in recent decades across academic, governmental, and applied settings (Caporaso and Presser 2024).

Yet the very success of cognitive interviewing has also drawn attention to its limitations. Although it offers valuable insight into individual- and subgroup-level variation in question validity (Collins 2015), it is also labour-intensive, slow, and expensive to implement. Conducting interviews, training interviewers, analysing recordings or notes, and iteratively revising items all require substantial resources, which means that only a limited number of questions, respondents, and design variants can usually be examined within real project constraints (Beatty and Willis 2007; Caporaso and Presser 2024). In practice, many studies rely on small convenience samples and a relatively small number of rounds, often because budgets and timelines do not permit more expansive testing (Beatty and Willis 2007). This creates a tension between the aspiration to identify as many important problems as possible and the reality that questionnaire development often proceeds under severe operational pressure.

That tension is exacerbated by evidence on how sample size relates to problems identified. Cognitive interviewing has often been justified on the basis that a small number of interviews can reveal the most serious problems, and in practice small samples are the norm: Caporaso and Presser (2024) report a median minimum of around five respondents and a median maximum of around twenty per round among US academic organisations. Yet consequential problems may remain undiscovered unless sample sizes become much larger than these conventional rules of thumb imply, because the probability of observing

a problem depends on both its prevalence and the interview’s sensitivity to it (Blair and Conrad 2011). For long or complex questionnaires, exhaustive human cognitive testing of every item is often infeasible.

Cognitive interviewing is better suited to problems respondents experience while answering than to unnoticed defects such as presuppositions or leading framing (Presser and Blair 1994). Systematic expert review instead applies methodological judgement and problem taxonomies directly to question text (Presser et al. 2004). The approaches tend to identify different kinds of problems, so neither is entirely sufficient on its own (Presser and Blair 1994), both also depend on scarce expertise and are therefore difficult to apply exhaustively across large instruments.

These long-standing constraints on effective question pre-testing have become newly salient with the introduction of large language models (LLMs), and their use in simulating human survey respondents, what is referred to as ‘silicon sampling’ (Argyle et al. 2023). This creates a new possibility for questionnaire design of using simulated respondents not to substitute for real populations, but to expose problems in draft questions. While early studies showed that conditioned models could reproduce aggregate attitudinal patterns and classic experimental effects (Argyle et al. 2023; Aher, Arriaga and Kalai 2023; Dillion et al. 2023; Sarstedt et al. 2024), subsequent evaluations have identified persistent limitations such as sensitivity to superficial ordering and label effects (Dominguez-Olmedo, Hardt and Mendler-Dünner 2024), systematic biases toward Western, educated, liberal subgroups (Atari et al. 2023; Santurkar et al. 2023), poor recovery of subgroup patterns in non-English settings (Qi, Lyu and Luo 2024; Heyde, Haensch and Wenz 2025), and a tendency for demographic prompting to flatten within-group diversity (Wang, Morgenstern and Dickerson 2025). These representational shortcomings currently make silicon sampling unsuitable for population inference. They are less consequential, however, in diagnostic applications where the goal is not to estimate population quantities but to identify question design flaws (Sarstedt et al. 2024). It is this diagnostic use case that the present paper addresses.

We present a systematic evaluation of LLM-based question-review procedures modelled on cognitive testing and expert review. For the cognitive-testing approach, we develop a three-stage pipeline in which synthetic respondents defined by real sociodemographic backstories work through questions in a think-aloud format in the first person, with an ‘analyst’ stage that codes identified problems and a synthesis stage that consolidates findings. We test three variants that differ in how much structure and guidance is provided in the prompt. As a comparison, we also evaluate direct LLM-based expert review, in which synthetic reviewers evaluate questions against known problem taxonomies. We again compare three prompt variants across expert review which vary in the level of structure and guidance. We assess performance using a test set of 27 questions, 20 of which contain deliberately embedded problems drawn from eight common problem types and seven of which are well-tested core items from the European Social Survey that serve as controls. We further test whether findings generalise across LLMs by replicating all configurations on two open-weight alternatives alongside the proprietary GPT-4o, and the newer proprietary Claude Opus 5. This allows us to evaluate discrimination between flawed and well-designed items, variation in the types of problems detected, and sensitivity to prompt design and LLM choice. It does not test whether any configuration performs comparably to human cognitive interviewers or expert reviewers.

Our contribution is a controlled evaluation in which detection and false-positive rates can be computed against a constructed answer key. The best performing configuration in the main run detects three quarters of known problems, although performance varies with prompts and LLM choice. Because complete pipeline configurations differ in several respects, contrasts cannot isolate an effect of cognitive testing versus expert review. The results support LLM-based review as a potential complement to human pretesting, particularly for early screening or where resources preclude intensive human evaluation.

2 Methodology

2.1 Simulated cognitive testing

To simulate cognitive testing, we developed an LLM-based pipeline for question evaluation, consisting of simulated respondents, analyst coding and synthesis. The architecture is model-agnostic and not tied to any specific interface. The simulated respondents are used to expose questions to diverse contextual circumstances in which different design problems may become salient. We do not consider them to be validated simulations of how real people with particular demographic characteristics would interpret or answer questions.

The pipeline proceeds in three stages (Figure 1). First, a random sample of n synthetic respondents is drawn from a pool of backstories derived from individual respondent records drawn from a representative sample of the target population. Here we use the 2018 European Social Survey (ESS) UK dataset, a probability sample of the UK general population aged 15 and over. The pool contains 2,204 respondent backstories. Each backstory is constructed deterministically by inserting values from the ESS record into a natural-language template: the backstories are not themselves generated by an LLM, so there is no feedback loop from the model used for cognitive testing back into the respondent profile.

Each sampled respondent works through the question in a single-prompt think-aloud format. An LLM ‘analyst’ then codes transcript-evidenced problems, and a synthesis step consolidates overlapping findings across respondents. Full prompts and coding requirements are provided in Appendix C. We use $n = 30$ because preliminary testing indicated that the number of identified problems tails off above this sample size; each run draws a different random backstory sample. Synthetic respondents are also prompted to select an answer and to rate their confidence in it from 1 to 5.

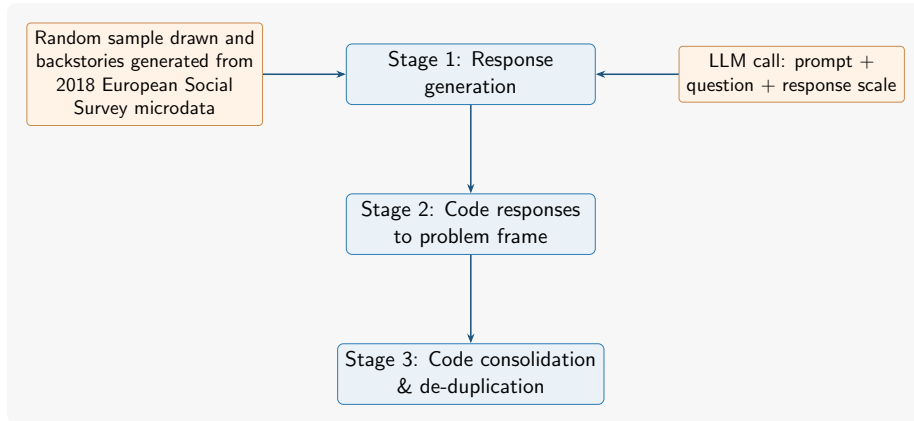


Figure 1: Simulated cognitive testing pipeline. Stage 1 draws a random sample of backstories from survey microdata and generates think-aloud transcripts. Stage 2 codes each transcript for problems. Stage 3 consolidates findings across respondents.

This architecture is modelled on selected procedural features of human cognitive interviewing: participants verbalise their thought process while answering a draft question and an analyst identifies patterns across interviews. This procedural resemblance should not be taken to imply that an LLM-generated think-aloud reproduces a human respondent’s cognitive processes or that the pipeline performs comparably to trained practitioners. An important simplification is that real cognitive interviews are dynamic: the interviewer follows up on the respondent’s answer with targeted probes, pursuing lines of difficulty as they emerge. In practice, however, many follow-up probes are planned in advance as part of the interview protocol and tailored to the question being tested. Fully modelling dynamic probing would require a multi-turn exchange between an LLM interviewer and an LLM respondent, which is beyond the scope of this initial investigation. The pipeline therefore relies on a single prompt to the respondent, raising the question of how much structure this prompt should contain.

We test this systematically with three prompt variants, holding all other content constant. In the first (CI-Open), the respondent simply thinks aloud with no framework or problem types mentioned. In the second (CI-Structured), the respondent works through the four-stage model of survey response developed by Tourangeau, Rips and Rasinski (2000) (comprehension, retrieval, judgement, response mapping) but no problem types are named. In the third (CI-List), the same four-stage structure includes explicit lists of common problem types at each stage (e.g., double-barrelled wording at comprehension, recall difficulty at retrieval, poor fit at response mapping). All three share the same analyst and synthesis prompts; full prompt texts are provided in Appendix C.

2.2 Expert review

For each question, the same LLM is prompted separately three times to produce three independent expert reviews. Each review identifies problems and assigns them a severity rating from 1 (negligible distortion)

to 10 (the answer is rendered meaningless). A synthesis stage then consolidates overlapping findings and averages their severity ratings across reviewers. As with cognitive interview, this architecture is modelled on expert appraisal but is not a validated simulation of human expert judgement. ER-Checklist evaluates seven problem types, including comprehension, response mapping, and presupposition. ER-Open uses no predefined types, while ER-Restrained adds instructions to report only consequential problems and return an empty list for a well-designed question. These are single-run configurations with full prompts provided in Appendix C.

2.3 Test set and evaluation design

The test set comprises 27 survey questions. Adapting the construction-based logic used by Blair and Conrad (2011), the first author introduced one to three established design problems into each of 20 conventional questions (Willis 2005; Presser and Blair 1994). These formed a deliberately non-exhaustive answer key, so unmatched findings could still be genuine problems. Seven well-tested European Social Survey (ESS) core items serve as controls.¹ The full set is provided in Appendix B.

This design allows us to assess two distinct aspects of performance. Detection rate measures the proportion of the 48 known problems that the pipeline successfully identifies and is equivalent to sensitivity in conventional diagnostic-test terminology: $TP/(TP + FN)$. We also report the mean number of pipeline findings per question that are not matched to the answer key. We refer to these as provisional false positives because the answer key records only the deliberately embedded problems and is not assumed to be exhaustive. On flawed questions, they are findings beyond those deliberately embedded; on ESS questions, they are any problems reported at all. Both estimates are likely to overstate genuinely spurious output, since the flawed questions may contain additional genuine problems and the ESS items, while developed to the highest standards, cannot be assumed to be entirely problem-free. We return to this point in the Results, where independent human assessment is used to evaluate the unmatched findings.

The reported provisional false-positive statistic is not the conventional diagnostic false-positive rate, $FP/(FP + TN)$. A question does not provide a finite, enumerated set of all the problems it does not contain, so the benchmark has no defined denominator of true-negative problem instances. Conventional specificity, $TN/(TN + FP)$, and overall diagnostic accuracy therefore cannot be calculated. Comparisons of output on flawed and ESS control questions instead provide a descriptive indication of how well each pipeline distinguishes questions known to contain embedded problems from well-tested questions with no known problems.

The use of deliberately constructed items with known embedded problems is not intended to approximate the full complexity of real-world questionnaire development. Rather, it provides a necessary identification strategy because without a known answer key, it is not possible to distinguish true positives from false positives or to compute detection rates in a principled way. The present design should therefore be understood as a benchmark test of diagnostic capability under controlled conditions, analogous to test batteries used in psychometric validation, rather than a direct simulation of applied questionnaire development.

Pipeline and answer-key problems were classified pairwise as exact, approximate, or non-matches, with one-to-one assignment. The six primary GPT-4o runs were judged independently by Claude Opus 4.6 and GPT-5.4 at temperature 0; agreement was 94% (721/764), and the first author adjudicated 43 disagreements. For reasons of cost and researcher time, the dual-judge procedure was restricted to the primary GPT-4o runs; the Llama-3.3, Qwen3, and Opus 5 outputs were matched by GPT-5.4 alone. Temperature is a sampling parameter controlling output variability, with lower values producing more deterministic responses; generation used 0.7 to permit variation among simulated respondents and matching used 0 for consistency. The complete protocol is provided in Appendix E.

2.4 LLMs used

All six configurations are first tested using GPT-4o (OpenAI; temperature 0.7), the proprietary LLM selected for the main experiment and accessed via the OpenAI API. GPT-4o was chosen as the primary LLM because its cost per token at the time of testing made the full benchmark, which involves many thousands of API calls across replicates, configurations, and questions, feasible within a fixed compute

¹An eighth source item, C2 (social trust), was excluded because its anchors are not unambiguously bipolar, making it unsuitable as a problem-free control.

budget. The same configurations are then replicated on two open-weight LLMs to test whether the pipeline’s performance generalises beyond a single proprietary system.

Llama-3.3-70B-Instruct (Meta) is an open-weight LLM with 70 billion parameters, served locally via SGLang.² Llama-3.3 is instruction-tuned but has undergone less extensive alignment than GPT-4o.

Qwen3-32B (Alibaba) is an open-weight LLM with 32 billion parameters, served locally via SGLang on the same hardware.³

During revision of the paper in August 2026, we added one run of each configuration using Claude Opus 5 (Anthropic), a frontier proprietary LLM at that time. This extension reflects the rapid pace of model development: the LLMs included in the study became available and were tested at different points, and differ in their capabilities and benchmark performance. Repeating the complete programme of runs and human assessment with Opus 5 was not feasible. The Opus 5 API did not support the temperature setting used for the other LLMs, so thinking was disabled and the output-token limit increased to prevent truncation. Its outputs were matched by GPT-5.4 and were not included in the human assessment.

For each LLM, all pipeline stages (respondent simulation, analyst coding, synthesis, and expert review) use the same LLM throughout. This tests each LLM’s end-to-end capability rather than mixing LLMs across stages. Prompts, test set, and backstories are held constant across the four-LLM comparison, with the Opus 5 API-setting exceptions described above.

3 Results

To give a sense of what the pipeline produces, we present output from both testing approaches applied to question F5: “Overall, how satisfied are you with your terms of employment?” (scale: Completely satisfied / Very satisfied / Fairly satisfied / Not very satisfied / Not at all satisfied). This question contains three deliberately embedded problems: “terms of employment” is vague, the response scale is asymmetric (three positive options, two negative), and the question presupposes current employment.

Table 1 shows output from both approaches. For CI-Open, two of the 30 simulated respondents are shown; neither received any cognitive framework or problem-type cues. For ER-Checklist, one of the three independent reviewers is shown alongside the consolidated output.

²Llama-3.3-70B is run with AWQ 4-bit quantisation (approximately 36GB VRAM) on a single NVIDIA RTX 6000 Ada 49GB GPU.

³Qwen3-32B is run with FP8 quantisation. Qwen3 generates reasoning blocks by default; a wrapper suppresses these by appending a no-think instruction and stripping reasoning tags from responses.

Table 1: Example pipeline output for question F5, “Overall, how satisfied are you with your terms of employment?” Top panel: CI-Open (respondent transcript and analyst coding). Bottom panel: ER-Checklist (expert reviewer assessment and consolidated problem code).

Simulated cognitive testing (CI-Open)	
Respondent transcript	Analyst coding
<i>Respondent 2 (retired, former shopkeeper):</i> “As I read the question about my satisfaction with my terms of employment, I first think about my situation. Being retired and a former shopkeeper, I’m not currently employed in the traditional sense, so the question feels a bit out of place for me. However, I can think back to when I was actively running my shop. [...] Since I’m retired, none of these apply directly to me anymore.”	The respondent is retired and self-employed, making the question about “terms of employment” feel irrelevant. They had to reflect on past experiences and adapt the question to fit their unique situation, which could lead to inaccuracies. (Severity: 7/10)
<i>Respondent 8 (kitchen helper, female, 54):</i> “As I read the question [...] I start to think about my current job as a kitchen helper. I consider the hours I work, the pay I receive, and the conditions I work under. These are what I think about when I hear ‘terms of employment.’ [...] The phrasing ‘terms of employment’ could be a bit vague, but I interpret it as encompassing these aspects: pay, hours, and work conditions.”	The respondent found the term “terms of employment” to be vague, which might have led to some uncertainty about what aspects to consider when answering the question. (Severity: 5/10)
Expert review (ER-Checklist)	
Expert reviewer assessment	Consolidated problem code
<i>Comprehension</i> (severity 6): “The phrase ‘terms of employment’ may be interpreted differently by respondents, potentially including salary, benefits, work hours, or job security.”	(1) “Terms of employment” is ambiguous (mean severity 5.7; flagged by all 3 reviewers).
<i>Presupposition</i> (severity 5): “The question assumes that respondents have defined and clear terms of employment, which may not be true for all, such as temporary or freelance workers.”	(2) The question assumes defined employment terms (mean severity 5.0; flagged by 1 reviewer).

These examples are illustrative rather than representative: they show how each pipeline operates when it successfully identifies an embedded problem. In these cases, cognitive testing surfaced the problems through respondent contexts, whereas expert review identified them from the question structure.

Table 2 presents the main GPT-4o results, based on one run of each configuration. These provide the common comparison across the six pipelines. Because findings that do not match the deliberately constructed answer key are only provisionally classified as false positives, we subsequently examine a separate set of unmatched CI-List and ER-Checklist findings through assessment by two questionnaire-design specialists.

Table 2: Overall results across six testing methods on GPT-4o. Detection = proportion of 48 known problems matched (equivalent to sensitivity). FP/Q = mean provisional false positives (findings unmatched to the deliberately constructed answer key) per question. Det. (≥ 5) and FP/Q (≥ 5) = detection and mean provisional false positives per question after excluding findings with pipeline-assigned severity below 5. Confidence = mean respondent confidence (1–5); available for cognitive testing methods only.

Configuration	Prompt architecture	Set	Detection	Det. (≥ 5)	FP/Q	FP/Q (≥ 5)	Confidence
CI-List	Simulated respondent works through the four-stage cognitive model with an explicit list of common problem types provided at each stage.	Flawed	75%	65%	1.8	1.50	4.08
		ESS	—	—	1.6	1.00	4.44
ER-Checklist	Three expert reviewers evaluate the question against a seven-type problem checklist (comprehension, double-barrelled, retrieval, response mapping, social desirability, presupposition, tautology).	Flawed	73%	60%	1.4	1.05	—
		ESS	—	—	2.3	0.57	—
ER-Open	Three expert reviewers identify problems in their own terms, with no predefined problem types and no additional restraint language.	Flawed	69%	69%	1.4	1.40	—
		ESS	—	—	3.1	3.14	—
CI-Open	Simulated respondent thinks aloud while working through the question, with no cognitive framework or problem types mentioned in the prompt.	Flawed	60%	50%	1.4	1.00	4.21
		ESS	—	—	1.4	1.00	4.41
ER-Restrained	Three expert reviewers identify problems in their own terms, with strong restraint language instructing them to report only problems genuinely likely to affect respondents and to return an empty list if the question is well designed.	Flawed	48%	46%	0.8	0.55	—
		ESS	—	—	0.3	0.29	—
CI-Structured	Simulated respondent works through the four-stage cognitive model, but no problem types are named at any stage.	Flawed	31%	25%	0.6	0.50	4.28
		ESS	—	—	0.7	0.57	4.42

CI-List has the highest detection rate (75%; 1.6 provisional false positives per ESS question), followed by ER-Checklist (73%; 2.3) and ER-Open (69%; 3.1), whose high sensitivity comes with frequent control-set findings, indicating poor discrimination. CI-Open achieves 60% detection with a moderate provisional false-positive rate (1.4 per ESS question). ER-Restrained has far fewer unmatched findings than any other configuration (0.3 per ESS question) but at the cost of missing more than half the known problems. CI-Structured is the weakest configuration overall at 31%; the four-stage framework appears to constrain GPT-4o’s simulated respondents so that they navigate even flawed questions without reporting difficulties.

Retaining only findings with severity at least 5 generally reduces provisional false positives at a modest cost in detection, which falls by between 0 and 13 percentage points across configurations (CI-List 75% to 65%; ER-Open unchanged). ER-Checklist falls from 2.3 to 0.57 unmatched findings per ESS question, whereas ER-Open’s unmatched findings are virtually unchanged (3.14 versus 3.1). Mean severity is higher on flawed questions (5.87, SD=1.22, $n = 318$) than controls (5.23, SD=1.15, $n = 66$), a modest difference of 0.64 points ($t(97.36) = 4.04$, $p = .0001$). Severity therefore offers some prioritisation signal despite

compressed calibration.

The provisional false-positive rates reported above treat the answer key as exhaustive and the ESS questions as entirely problem-free, neither of which is likely to hold in practice. We therefore conducted a separate human assessment to examine whether unmatched findings can represent genuine problems missing from the answer key. Eighty unmatched findings were submitted to two questionnaire-design specialists, of which 79 received ratings from both and form the analysis reported here. Each specialist rated each finding on a 1–5 severity scale (1 = not a problem at all, 5 = very severe problem) without seeing the answer key or matching verdicts. Ratings were collapsed into a binary classification (4–5 = genuine design problem, 1–3 = not genuine), and a finding was classified as genuine only where both specialists independently rated it 4 or 5. Inter-rater agreement on the binary classification was 79% (Cohen’s $\kappa = 0.57$; Spearman $r_s = 0.71$ on the original 1–5 scale).

Among the 79 findings rated by both specialists, 19 of 29 unmatched CI-List findings on flawed questions (66%) and 8 of 24 ER-Checklist findings (33%) were classified as genuine problems. On ESS questions, the corresponding figures were 0 of 9 for CI-List and 2 of 17 for ER-Checklist. This separate assessment therefore supports two interpretive points: an answer key containing only deliberately embedded problems can misclassify plausible additional defects as false positives, while many unmatched findings on well-tested control questions are genuinely spurious. Because the assessment used different runs, these proportions should not be used to adjust the rates in Table 2; rather, they show why those rates are provisional and why human review remains necessary.

The simulated cognitive testing pipeline introduces stochastic variation between runs from both the random sampling of backstories and the inherent non-determinism of LLM text generation (the same prompt can produce different outputs across calls at temperature>0). Table 2 alone cannot show how much a result from a single model and configuration would vary if the run were repeated. To quantify this within-model variability, we ran each cognitive testing configuration three additional times with different random seeds, holding all other parameters constant. Table 3 shows the repeated-run results.

Table 3: Per-replicate results for the three cognitive testing configurations on GPT-4o ($n = 30$).

Configuration	Run	Detection	FP/flawed Q	FP/ESS Q
CI-List	Rep 1	79%	1.4	1.4
	Rep 2	67%	1.1	1.8
	Rep 3	65%	1.4	1.6
	<i>Mean (SD)</i>	<i>70% (7.9)</i>	<i>1.30</i>	<i>1.58</i>
	<i>Range</i>	<i>65–79</i>	<i>1.1–1.4</i>	<i>1.4–1.8</i>
CI-Open	Rep 1	58%	1.2	1.0
	Rep 2	58%	1.2	1.2
	Rep 3	60%	1.1	1.1
	<i>Mean (SD)</i>	<i>59% (1.2)</i>	<i>1.18</i>	<i>1.12</i>
	<i>Range</i>	<i>58–60</i>	<i>1.1–1.2</i>	<i>1.0–1.2</i>
CI-Structured	Rep 1	29%	0.4	0.2
	Rep 2	33%	0.5	0.9
	Rep 3	40%	0.5	0.4
	<i>Mean (SD)</i>	<i>34% (5.2)</i>	<i>0.45</i>	<i>0.50</i>
	<i>Range</i>	<i>29–40</i>	<i>0.4–0.5</i>	<i>0.2–0.9</i>

There is notable between-run variation, but the general ordering of detection and provisional false-positive rates is broadly consistent across repeated runs. CI-List achieves the highest mean detection (70%) but is also the most variable (standard deviation of 7.9 percentage points, henceforth pp), with individual estimates ranging from 65% to 79%. Its 75% primary estimate in Table 2 lies within this range but should not be read as a stable or uniquely representative performance level. CI-Open is considerably more stable (SD = 1.2 pp) but at a lower mean detection level of 59%; its 60% primary estimate is at the top of the repeated-run range. CI-Structured remains weak across repeated runs (mean 34%, range 29–40%), showing that its low primary estimate is not simply an anomalous single run. Provisional false-positive rates are more stable than detection rates across all three configurations. The repeated runs therefore support the broad comparative pattern in Table 2, while showing that single-run detection percentages, especially for CI-List, carry meaningful within-model uncertainty.

Detection rates by problem type

Table 4 shows that detection varies substantially, and different configurations have distinctive profiles. Double-barrelled problems are detected by all configurations except CI-Structured (67%), presumably because asking about two things at once is salient and well known. Presupposition is the hardest type overall and the one where configurations diverge most: ER-Checklist leads at 89%, likely because its checklist explicitly names presupposition as a category, while ER-Restrained manages only 11% without this cue.

The most substantively interesting contrast concerns response-mapping problems. CI-List detects 71% of these problems and CI-Open 57%, compared to 43% for ER-Checklist and 0% for ER-Restrained. In the cognitive-testing configurations, the synthetic respondent is explicitly required to select and discuss a response option, whereas the expert-review configurations analyse the question and response scale directly. This difference may help explain the observed detection profiles, but the design cannot identify an independent effect of cognitive testing versus expert review because pipeline architecture, prompts, and processing stages vary together.

Leading and loaded framing shows a large gap between configurations with and without explicit type cues. CI-Open detects only 40%, while CI-List reaches 100% and ER-Checklist 80%. Certain problem types, particularly presupposition and leading framing, are rarely identified without explicit prompting or taxonomic guidance.

There is a strong positive association between detection rate and mean severity across problem types (Spearman $\rho = 0.69$), such that problem types that the pipeline detects more frequently also receive higher severity ratings, with double-barrelled problems at one extreme and response-mapping at the other. This could indicate that genuinely severe problems are more salient and therefore easier to detect. Alternatively, problems detected by more respondents accumulate stronger evidence, which may inflate the synthesised severity rating. A different research design would be needed to disentangle these possibilities.

The confidence ratings elicited from simulated respondents in the cognitive testing configurations provide another potential diagnostic signal. Across all three CI configurations, simulated respondents report consistently lower confidence on flawed questions than on ESS controls (Table 2): CI-List 4.08 versus 4.44, CI-Open 4.21 versus 4.41, CI-Structured 4.28 versus 4.42. The direction of the difference is consistent, though the gap is modest in absolute terms. At the problem-type level (Table 4), the pattern is more differentiated. Problem types that create experienced difficulty for the respondent produce lower confidence. Leading/loaded framing (3.80) and presupposition (3.91) both fall well below the ESS baseline. At the question level, F14 (childcare satisfaction, which presupposes need for childcare) averages 2.10, and F4 (immigration attitudes, with leading preamble and complex syntax) averages 3.37. By contrast, problem types that affect data quality without the respondent noticing produce near-ceiling confidence: vague/ambiguous wording (4.37), retrieval/recall (4.32), and response-mapping (4.30) all fall close to the ESS control mean. Confidence ratings thus better identify problems that create experienced difficulty for the respondent, as would be expected.

Table 4: Detection rate (%) and mean LLM-assigned severity (1–10) by problem type across six testing methods on GPT-4o, using the dual-judge match matrix with first-author adjudication of disagreements. Severity is pooled across all methods for matched true positives. Confidence is the mean respondent rating (1–5) pooled across CI configurations only.

Problem type	N	Cognitive testing			Expert review			Severity	Confidence
		Open	Struct.	List	Checklist	Open	Restr.		
Double-barrelled	6	100	67	100	100	100	100	7.18	4.07
Presupposition	9	44	33	44	89	56	11	6.18	3.91
Vague/ambiguous	9	56	11	78	67	78	67	5.57	4.37
Response-mapping	7	57	43	71	43	29	0	5.33	4.30
Retrieval/recall	5	100	0	100	100	80	80	5.64	4.32
Leading/loaded	5	40	20	100	80	80	40	6.33	3.80
Sensitivity	4	25	50	50	50	50	50	6.07	4.29
Complex syntax	3	67	33	67	33	100	67	5.97	4.04

Finally, to test whether performance generalises beyond GPT-4o, we extended the comparison to two leading open-weight LLMs available at the time, Llama-3.3-70B-Instruct and Qwen3-32B, running on a local GPU server, and to Claude Opus 5, a current frontier proprietary LLM.⁴ Table 5 presents the full comparison.

Table 5: Four-LLM comparison across all six configurations. Det. = detection rate (sensitivity). FP/ESS = provisional false positives (unmatched findings) per ESS control question. The lower panel retains only findings with pipeline-assigned severity ≥ 5 . GPT-4o values use the dual-judge match matrix with adjudication; Llama-3.3, Qwen3, and Opus 5 values use GPT-5.4 matching. Opus 5 values are based on one run per configuration.

Config	GPT-4o		Llama-3.3-70B		Qwen3-32B		Opus 5	
	Det.	FP/ESS	Det.	FP/ESS	Det.	FP/ESS	Det.	FP/ESS
<i>All findings</i>								
CI-Open	60%	1.4	73%	5.4	79%	6.7	94%	9.7
CI-Structured	31%	0.7	65%	4.1	73%	5.4	92%	9.1
CI-List	75%	1.6	88%	5.4	75%	7.1	88%	8.9
ER-Checklist	73%	2.3	79%	3.4	73%	2.7	100%	8.4
ER-Open	69%	3.1	67%	3.1	63%	1.7	100%	8.4
ER-Restrained	48%	0.3	46%	0.3	4%	0.1	94%	5.3
<i>Severity ≥ 5</i>								
CI-Open	50%	1.0	73%	4.7	44%	2.1	60%	2.0
CI-Structured	25%	0.6	56%	3.9	46%	0.7	52%	1.1
CI-List	65%	1.0	79%	4.4	56%	3.1	63%	1.7
ER-Checklist	60%	0.6	65%	1.9	33%	0.4	83%	1.4
ER-Open	69%	3.1	67%	3.0	56%	1.4	79%	3.7
ER-Restrained	46%	0.3	44%	0.1	4%	0.1	60%	0.9

Table 5 shows that higher detection is often accompanied by more unmatched findings. This is clearest for the cognitive-testing configurations, where GPT-4o generally reports the fewest problems and has the lowest detection, while Opus 5 reports the most and has the highest detection. Llama and Qwen3 tend to fall between them. The higher detection achieved by more expansive LLMs must therefore be interpreted partly in relation to the volume of problems they report: generating more candidate problems creates more opportunities to match the answer key. Thus, although Opus 5 detects every embedded problem under two expert-review configurations, it also reports 8.4 unmatched findings per control question. The relationship is not universal, however. For ER-Open, the open-weight LLMs produce no more unmatched findings than GPT-4o and have slightly lower detection.

Severity ratings provide a useful filter for some LLMs but not for others. For Opus 5, applying a threshold of 5 reduces ER-Checklist’s unmatched findings from 8.4 to 1.4 per control question while retaining detection of 83% of the embedded problems. The same threshold has little effect on Llama’s cognitive-testing outputs because it assigns high severity to most findings. For Qwen3, by contrast, thresholding removes many genuine detections as well as unmatched findings; ER-Checklist detection falls from 73% to 33%. Nor does restraint within the prompt operate consistently across LLMs. ER-Restrained produces relatively few unmatched findings on GPT-4o and Llama, almost entirely suppresses output on Qwen3, but has little restraining effect on Opus 5. ER-Checklist shows the most consistent raw detection across LLMs, ranging from 73% to 100%, which suggests that supplying a fixed taxonomy may reduce some of the variation arising from LLM choice.

These comparisons nevertheless remain exploratory. Each cell is based on a single run, the non-GPT-4o outputs were matched by one LLM judge, and the prompts were developed on GPT-4o rather than optimised separately for each LLM. Opus 5 also required different API settings. As Table 3 shows, repeated runs of the same configuration can differ appreciably even when the LLM and prompt are unchanged. The cross-LLM results are therefore most informative about broad patterns, rather than small differences between individual percentages.

⁴The open-weight runs were conducted in March 2026. Opus 5 was run in August 2026 while the submitted manuscript was being revised, by which time the proprietary frontier had advanced. “Current” therefore refers to the leading LLMs available at the time of each extension rather than to a single common date.

4 Discussion

Cognitive interviewing and expert review have become indispensable components of the survey development workflow, providing important means by which researchers identify design flaws in draft questions before they reach the field. Yet both remain significant bottlenecks. They require scarce expertise, are time-consuming and expensive, and in practice can cover only a fraction of the items in a long or complex questionnaire. The practical consequence is that many questions go to field with less testing than would be ideal, and some go with none at all. LLM-based procedures modelled on selected features of these methods could extend diagnostic coverage to contexts where intensive human pretesting is currently out of reach. Whether they reproduce the judgements or performance of human practitioners is an interesting but separate question that the present study is not able to assess.

The results of this study suggest that the potential of LLMs for detecting problems in draft questions is real, if uneven. The best-performing simulated cognitive testing configuration, CI-List, detected three quarters of the 48 known problems in the primary analysis. Across three repeated runs, however, it averaged 70% and ranged from 65% to 79%. The primary result should therefore be understood as one realisation of a stochastic pipeline rather than a fixed estimate of its performance. CI-Open was more stable across runs but detected fewer problems, while CI-Structured remained weak on GPT-4o throughout. In a separate assessment using outputs from additional runs, two questionnaire-design specialists classified many findings unmatched to the deliberately constructed answer key as genuine design problems. This shows that the answer key is not exhaustive and that unmatched findings require human interpretation.

The cognitive-testing and expert-review configurations produced somewhat different detection profiles on this benchmark. The cognitive-testing configurations were more effective at identifying response-mapping failures, whereas the expert-review configurations were stronger on presupposition and leading framing. These contrasts are consistent with the procedural emphasis of the respective pipelines, but cannot be interpreted as evidence that human cognitive interviewing and expert appraisal would show the same pattern. This result also supports a practice of combining outputs from differently designed LLM pipelines as a practical strategy.

Detection varied substantially by problem type. Double-barrelled and retrieval problems were detected readily, whereas CI-Open found only 44% of presuppositions and no configuration exceeded 50% for sensitivity problems. Two improvement routes follow: prompts can cue text-visible defects, while larger or purposively selected respondent pools may expose circumstance-dependent problems such as inapplicability.

The comparison across LLMs shows that higher detection often comes at the cost of reporting more unmatched findings. This pattern is clearest for the cognitive-testing configurations: GPT-4o generally reports the fewest problems and detects the fewest embedded defects, while Opus 5 reports and detects the most, with Llama and Qwen3 usually falling between them. Some of the apparent advantage of the more expansive LLMs is therefore mechanical, since generating more candidate problems creates more opportunities to match the answer key. Raw detection alone is consequently an incomplete guide to practical value. A useful question-review procedure must also discriminate between questions that warrant revision and those that do not.

The severity-thresholded results provide a more discriminating comparison. Here the strongest result is produced by the model closest to the proprietary frontier at the time of writing: Opus 5 with ER-Checklist retains 83% detection while reducing unmatched findings from 8.4 to 1.4 per control question. Yet severity is not a uniformly reliable filter. Llama assigns high severity to most of its findings, so thresholding removes little of its excess output, whereas Qwen3 assigns low severity to many genuine detections and loses substantial sensitivity. The restraint instruction is similarly model-dependent: it gives the intended conservative behaviour on GPT-4o and Llama, almost silences Qwen3, and has relatively little restraining effect on Opus 5. By contrast, ER-Checklist maintains comparatively high raw detection across all four LLMs, suggesting that a fixed taxonomy may make the task less sensitive to differences between them.

These comparisons do not, however, constitute controlled tests of the LLMs' underlying capabilities. The prompts were developed on GPT-4o and applied elsewhere without model-specific optimisation; the CI-Structured reversal across LLMs illustrates how consequential that interaction can be (see Appendix D). Each comparison cell is also based on one run, the non-GPT-4o outputs were matched by a single judge, and Opus 5 required different API settings. The results therefore show that performance depends

strongly on both LLM choice and calibration of the prompt and severity scale. They also demonstrate why conclusions about the “best” available LLM will be time-bound in a field developing as rapidly as this one.

Questions containing presuppositions that made them clearly inapplicable, or leading framing that created discomfort, produced markedly lower respondent confidence. Questions whose problems were subtler, involving vague terminology or recall difficulty, did not. This pattern maps onto a well-established distinction in the cognitive interviewing literature between problems that create experienced difficulty for the respondent and problems that silently degrade measurement quality without the respondent noticing (Presser and Blair 1994). Confidence ratings may therefore provide a useful screening signal, flagging questions where simulated respondents sense something is wrong, but they cannot substitute for explicit problem identification.

A 20-item simulated cognitive-testing evaluation costs under \$10 and completes in under an hour on GPT-4o; expert review costs under \$1 and completes in minutes. Both are substantially cheaper and faster than human pretesting; call-level details are provided in Appendix F.

Several limitations of the present study warrant acknowledgement. The test set comprised deliberately constructed items with known embedded problems, which provides the identification strategy needed to compute detection rates but limits external validity. The embedded problems are by design more clearly defined than many issues encountered in real questionnaire development, where defects may be subtle, overlapping, or context-dependent. Reported detection rates are therefore likely to represent an upper bound on what would be achieved with naturally occurring draft questions. Evaluating LLM-based approaches on organically generated questionnaires is an important direction for future work, but would require alternative evaluation frameworks that do not rely on a fixed answer key.

There is also a limitation to exact computational reproduction of the repeated-run analysis. The complete archived outputs for the nine runs summarised in Table 3 were retained on a local GPU server but appear to have been deleted during a subsequent server upgrade and restart. The configurations, seeds, and reported summary values were retained. Re-running the pipelines would not reconstruct the original evidence: the outputs are stochastic, GPT-4o was accessed through a moving API model identifier, and the available proprietary and open-weight models have subsequently changed. Such runs would therefore constitute a new temporal and cross-model replication, potentially requiring a broader update of the paper, rather than exact reproduction of Table 3. We retain the reported values and disclose this shortfall explicitly.

The study includes no human baseline: practitioners did not evaluate the same 27 questions. The human pretesting literature provides conceptual motivation and an interpretive benchmark, not an empirical comparison within this study. The 75% detection rate therefore cannot be interpreted as approaching, matching, or exceeding human performance; that comparison would require human review of the identical set under a prespecified protocol.

The single-prompt pipeline omits the dynamic probing central to human cognitive interviewing. Its backstory-conditioned outputs are diagnostic scenarios, not evidence about how people with those characteristics actually respond. Future work could test multi-turn probing and vary the backstory pool, temperature, and respondent sample size.

Using more than 30 backstories would increase the probability of including uncommon respondent circumstances in which otherwise latent applicability or interpretation problems become salient, and might therefore increase the number of problems detected. Whether this would improve the balance between detection and unmatched findings remains an empirical question.

These findings are an initial benchmark in a rapidly changing LLM landscape. Repeated runs remain important because single realisations can misstate performance. LLM review is unlikely to replace skilled cognitive interviewing, but its speed and low cost create a practical opportunity to broaden and repeat diagnostic screening where human pretesting is constrained.

In practical use, LLM-based question review is best positioned as an early-stage screening and diagnostic tool. It may be particularly useful when researchers need to review long instruments, compare successive drafts, or examine more questions than can feasibly receive intensive human pretesting. The results suggest that researchers should use more than one prompting approach where possible, because differently designed pipelines identify complementary problems, and should treat the resulting findings as candidates for investigation rather than automatic instructions to revise a question. The outputs can help researchers

prioritise questions and issues for closer examination, but decisions about whether and how to revise an item remain substantive and methodological judgements.

Human cognitive interviewing remains necessary when the aim is to understand how real respondents interpret a question, retrieve information, form judgements, or use the response options, particularly for sensitive, culturally dependent, or high-stakes measures. Human experts are also needed to assess construct validity, questionnaire context, and trade-offs between competing design principles, and to adjudicate plausible problems not anticipated by a benchmark or prompt. A productive workflow would therefore use LLM review to broaden initial diagnostic coverage and support iterative drafting, followed by targeted human expert review and cognitive testing for the questions and problems that matter most.

References

- Aher, Gati V., Rosa I. Arriaga and Adam Tauman Kalai (3rd July 2023). ‘Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies’. In: *Proceedings of the 40th International Conference on Machine Learning*. International Conference on Machine Learning, PMLR, pp. 337–371. URL: <https://proceedings.mlr.press/v202/aher23a.html> (visited on 14/10/2025).
- Argyle, Lisa P. et al. (July 2023). ‘Out of One, Many: Using Language Models to Simulate Human Samples’. In: *Political Analysis* 31.3, pp. 337–351. ISSN: 1047-1987, 1476-4989. DOI: 10.1017/pan.2023.2. URL: <https://www.cambridge.org/core/journals/political-analysis/article/out-of-one-many-using-language-models-to-simulate-human-samples/035D7C8A55B237942FB6DBAD7CAA4E49> (visited on 09/10/2023).
- Atari, Mohammad et al. (22nd Sept. 2023). *Which Humans?* DOI: 10.31234/osf.io/5b26t. URL: https://osf.io/5b26t_v1 (visited on 08/10/2025).
- Beatty, Paul C. and Gordon B. Willis (1st Jan. 2007). ‘Research Synthesis: The Practice of Cognitive Interviewing’. In: *Public Opinion Quarterly* 71.2, pp. 287–311. ISSN: 0033-362X. DOI: 10.1093/poq/nfm006. URL: <https://doi.org/10.1093/poq/nfm006> (visited on 19/12/2025).
- Blair, Johnny and Frederick G. Conrad (2011). ‘Sample Size for Cognitive Interview Pretesting’. In: *The Public Opinion Quarterly* 75.4, pp. 636–658. ISSN: 0033-362X. URL: <https://www.jstor.org/stable/41288411> (visited on 10/12/2025).
- Caporaso, Andrew and Stanley Presser (1st Nov. 2024). ‘The Prevalence and Nature of Cognitive Interviewing as a Survey Questionnaire Evaluation Method in the United States’. In: *Journal of Survey Statistics and Methodology* 12.5, pp. 1278–1294. ISSN: 2325-0992. DOI: 10.1093/jssam/smad047. URL: <https://doi.org/10.1093/jssam/smad047> (visited on 19/12/2025).
- Collins, Debbie (2015). *Cognitive Interviewing Practice*. SAGE Publications Ltd. ISBN: 978-1-4739-1010-2. DOI: 10.4135/9781473910102. URL: <https://methods.sagepub.com/book/edvol/cognitive-interviewing-practice/toc> (visited on 14/04/2026).
- Dillion, Danica et al. (1st July 2023). ‘Can AI language models replace human participants?’ In: *Trends in Cognitive Sciences* 27.7, pp. 597–600. ISSN: 1364-6613. DOI: 10.1016/j.tics.2023.04.008. URL: <https://www.sciencedirect.com/science/article/pii/S1364661323000980> (visited on 26/09/2025).
- Dominguez-Olmedo, Ricardo, Moritz Hardt and Celestine Mendler-Dünnér (9th Dec. 2024). *Questioning the Survey Responses of Large Language Models*. DOI: 10.48550/arXiv.2306.07951. arXiv: 2306.07951[cs]. URL: <http://arxiv.org/abs/2306.07951> (visited on 16/03/2026).
- Heyde, Leah von der, Anna-Carolina Haensch and Alexander Wenz (26th Apr. 2025). ‘Vox Populi, Vox AI? Using Large Language Models to Estimate German Vote Choice’. In: *Social Science Computer Review*, p. 08944393251337014. ISSN: 0894-4393. DOI: 10.1177/08944393251337014. URL: <https://doi.org/10.1177/08944393251337014> (visited on 26/09/2025).
- Presser, Stanley and Johnny Blair (1994). ‘Survey Pretesting: Do Different Methods Produce Different Results?’ In: *Sociological Methodology* 24, pp. 73–104. ISSN: 0081-1750. DOI: 10.2307/270979. URL: <https://www.jstor.org/stable/270979> (visited on 09/01/2026).
- Presser, Stanley et al., eds. (2004). *Methods for Testing and Evaluating Survey Questions*. Section: 1 eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/0471654728.ch1>. John Wiley & Sons, Ltd. ISBN: 978-0-471-65472-8. DOI: 10.1002/0471654728.ch1. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/0471654728.ch1> (visited on 14/04/2026).
- Qi, Weihong, Hanjia Lyu and Jiebo Luo (16th July 2024). *Representation Bias in Political Sample Simulations with Large Language Models*. DOI: 10.48550/arXiv.2407.11409. arXiv: 2407.11409[cs]. URL: <http://arxiv.org/abs/2407.11409> (visited on 26/09/2025).

- Santurkar, Shibani et al. (3rd July 2023). ‘Whose Opinions Do Language Models Reflect?’ In: *Proceedings of the 40th International Conference on Machine Learning*. International Conference on Machine Learning. PMLR, pp. 29971–30004. URL: <https://proceedings.mlr.press/v202/santurkar23a.html> (visited on 08/10/2025).
- Sarstedt, Marko et al. (2024). ‘Using large language models to generate silicon samples in consumer and marketing research: Challenges, opportunities, and guidelines’. In: *Psychology & Marketing* 41.6. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/mar.21982>, pp. 1254–1270. ISSN: 1520-6793. DOI: 10.1002/mar.21982. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/mar.21982> (visited on 26/09/2025).
- Tourangeau, Roger, Lance J. Rips and Kenneth Rasinski (2000). *The Psychology of Survey Response*. Cambridge: Cambridge University Press. DOI: 10.1017/CB09780511819322.
- Wang, Angelina, Jamie Morgenstern and John P. Dickerson (Mar. 2025). ‘Large language models that replace human participants can harmfully misportray and flatten identity groups’. In: *Nature Machine Intelligence* 7.3, pp. 400–411. ISSN: 2522-5839. DOI: 10.1038/s42256-025-00986-z. URL: <https://www.nature.com/articles/s42256-025-00986-z> (visited on 26/09/2025).
- Willis, Gordon (2005). *Cognitive Interviewing*. 2455 Teller Road, Thousand Oaks California 91320 United States of America: SAGE Publications, Inc. ISBN: 978-0-7619-2804-1 978-1-4129-8365-5. DOI: 10.4135/9781412983655. URL: <https://methods.sagepub.com/book/cognitive-interviewing> (visited on 18/03/2026).

Data availability statement

An anonymised replication package is available during peer review at [ANONYMOUS OSF URL]. It contains the benchmark, prompts, configurations, retained outputs, matching and adjudication materials, anonymised human ratings, and analysis code. The original files underlying Table 3 were lost during a local server upgrade; retained configurations, seeds, and summaries are supplied, but stochastic re-runs on subsequently changing LLMs would not reconstruct them. The repository will be made public with a permanent citation on publication.

Appendix A AAPOR-Required Disclosure Elements

The data analysed in this paper are primarily LLM-generated outputs produced by the simulated cognitive testing and expert review pipelines described in the Methodology. These data do not fit the canonical frame of survey research on human respondents, and several AAPOR disclosure elements therefore do not apply. We list all elements below, with an explanation where a given element is not applicable. The paper also makes secondary use of microdata from the 2018 European Social Survey (UK), which is documented as a second data source.

Data source 1: LLM-generated question-evaluation outputs

Type and source of data: New data collected by the authors. Outputs were generated by prompting large language models via API and recording their responses. No human respondents were involved in producing these data.

Data Collection Strategy: Automated prompting of LLMs through a Python-based pipeline. For simulated cognitive testing, each test question was evaluated in three stages: (i) $n = 30$ simulated respondents each produced a think-aloud transcript in response to the question, conditioned on a demographic backstory; (ii) an analyst LLM coded each transcript for evidence of question-design problems; (iii) a synthesis LLM consolidated findings across the 30 respondents. For expert review, three LLM reviewers evaluated each question directly, followed by a synthesis step. Six prompt configurations (three cognitive testing, three expert review) were run across four LLMs.

Research Sponsor and Conductor: The research was conducted by the authors at their respective institutions and funded by a national research council [name withheld for review].

Measurement Tools/Instruments: All prompt texts used in the pipeline are provided verbatim in Appendix C. The full test set of 27 questions (20 deliberately flawed items and 7 ESS control items) is provided in Appendix B.

Population Under Study: Not applicable in the conventional sense. The unit of analysis is the survey question ($n = 27$), not a population of respondents. LLM outputs conditioned on 2,204 synthetic respondent backstories (see Data Source 2) serve as the source of variation in simulated cognitive testing.

Methods Used to Generate and Recruit the Sample: Not a probability sample in the conventional sense. For each simulated cognitive testing run, 30 backstories were drawn by simple random sampling without replacement from the pool of 2,204 backstories derived from the 2018 ESS UK (see Data Source 2). Expert review uses three fixed reviewers per question with no backstory sampling. No quotas were used. No human participants were recruited; no incentives were provided.

Method(s) and Mode(s) of Data Collection: API calls to four large language models, all prompts and responses in English:

- GPT-4o (OpenAI), proprietary, accessed via the OpenAI API at temperature 0.7.
- Llama-3.3-70B-Instruct (Meta), open-weight, served locally via SGLang with AWQ 4-bit quantisation on a single NVIDIA RTX 6000 Ada 49GB GPU.
- Qwen3-32B (Alibaba), open-weight, served locally via SGLang with FP8 quantisation on the same hardware.
- Claude Opus 5 (Anthropic), proprietary, accessed via the Anthropic API for one run per configuration, with thinking disabled and no temperature parameter.

For the six primary GPT-4o runs, matching of pipeline-identified problems to the answer key was performed independently by Claude Opus 4.6 (Anthropic) and GPT-5.4 (OpenAI) at temperature 0, with disagreements adjudicated by the first author. The Llama-3.3, Qwen3, and Opus 5 outputs were matched by GPT-5.4 alone. The matching judge for the repeated GPT-4o runs cannot be verified from the retained files.

Dates of Data Collection: Not applicable in the conventional sense. The study does not rely on a fixed fieldwork period; pipeline runs were executed across the course of the project and can be re-run at any time given the prompts, test set, and model identifiers documented here.

Sample Sizes and Precision: The unit of analysis is the question ($n = 27$: 20 flawed, 7 ESS controls). For each cognitive testing configuration and LLM, 30 simulated respondents were generated per question. For each expert review configuration and LLM, three reviewer outputs were generated per question. The comparison comprises six configurations $\times$ four LLMs $\times$ 27 questions, with one run per configuration–LLM combination; GPT-4o cognitive-testing configurations also have three additional runs. No sampling-error estimates are reported: the study is not a sample survey of a population, and reported detection and provisional false-positive rates are descriptive summaries of pipeline behaviour on a fixed question set. Inter-judge agreement for primary-run answer-key matching was 94% across 764 problem pairs evaluated by the two LLM judges (Claude Opus 4.6 and GPT-5.4); the 43 discordant pairs were adjudicated by the first author.

Whether and How the Data Were Weighted: Not applicable. Data were not weighted.

How the Data Were Processed and Procedures to Ensure Data Quality: Prompts, test set, and backstories were held constant across the four-LLM comparison, subject to the documented Opus 5 API-setting exceptions. Identical prompt texts were used for every question within a configuration. Coding of pipeline outputs into problem reports was performed by the LLMs themselves under the analyst and synthesis prompts provided in Appendix C. Primary-run matching was performed by two independent LLM judges with disagreements resolved by the first author; non-primary matching provenance is stated above. Pipeline outputs were stored as structured records (JSON) with full transcripts retained for audit, except for the repeated-run files whose loss is disclosed in the manuscript.

Panel Description: Not applicable. No human or persistent-agent panel was used.

Interviewer Details: Not applicable. No human interviewers were involved. The simulated interviewer role is performed by the LLM under the think-aloud prompts documented in Appendix C. The LLM judges used for answer-key matching are described above.

Screening Criteria and Process: Not applicable. No human respondents were screened.

Study Stimuli: The 27 test questions (20 deliberately flawed items and 7 ESS control items) constitute the study stimuli and are reproduced in full in Appendix B.

Dispositions or Response or Participation Rates: Not applicable. LLM API calls do not have meaningful dispositions or non-response. All scheduled calls in the benchmark returned usable output; isolated API errors were retried.

Measurement and Model Specification: The pipeline architecture is fully described in the Methodology section and the prompts are reproduced in Appendix C. Replication code, the full test set and answer key, and archived pipeline outputs are available at the repository listed in the Data Availability Statement.

Limitations: Reported detection rates are based on a test set of deliberately constructed items with known embedded problems and therefore represent an upper bound on what would be achieved with naturally occurring draft questions. The pipeline relies on a single-prompt think-aloud rather than multi-turn dynamic probing. Temperature, number of simulated respondents, and backstory content were held constant; other settings are likely to affect results. These limitations are discussed in the Discussion section.

Data source 2: 2018 European Social Survey (United Kingdom), used to construct synthetic respondent backstories

Type and source of data: Secondary use of existing survey microdata. Used only to construct natural-language backstories for simulated respondents; no substantive analysis of ESS data is reported.

Data Collection Strategy: Face-to-face probability-sample survey of the UK adult population, conducted as part of Round 9 of the European Social Survey.

Research Sponsor and Conductor: The European Social Survey ERIC (ESS ERIC). The UK fieldwork in Round 9 was conducted by NatCen Social Research. Full methodological documentation is available at <https://www.europeansocialsurvey.org>.

Measurement Tools/Instruments: The variables used to construct backstories were age, gender, education, employment status, region, household composition, and income band. The full ESS Round 9

questionnaire is published on the ESS website.

Population Under Study: Residents aged 15 and over in private households in the United Kingdom.

Methods Used to Generate and Recruit the Sample: Probability sample. Design, frame, and fieldwork procedures are documented in the ESS Round 9 methodology reports published by ESS ERIC.

Method(s) and Mode(s) of Data Collection: Computer-assisted personal interviewing (CAPI) in English.

Dates of Data Collection: 2018–2019 (ESS Round 9 UK fieldwork).

Sample Sizes: 2,204 respondents, each contributing one backstory to the pool.

Whether and How the Data Were Weighted: Not applicable for the present paper. Backstories are drawn by simple random sampling from the 2,204 records without weighting. The authors' use of ESS microdata is not intended to produce population estimates.

How the Data Were Processed and Procedures to Ensure Data Quality: ESS data are processed and quality-checked by ESS ERIC prior to public release. The authors applied no additional editing beyond selecting the variables listed above and inserting them deterministically into a natural-language template.

Panel Description, Interviewer Details, Screening Criteria, Study Stimuli, Dispositions: These elements relate to original ESS fieldwork, documented in the ESS methodology reports, and are not reproduced here.

Measurement and Model Specification: Not applicable. ESS data are used only as inputs to backstory construction, not in any statistical model.

Limitations: The backstories are simplified profiles derived from selected ESS variables and do not reproduce the full circumstances or response processes of ESS respondents. Representativeness of the UK population is not a property the pipeline requires, since backstories serve to diversify the simulated respondent pool rather than to generate population estimates.

Appendix B Test Set

Appendix B.1 Flawed questions (n = 20, 48 known problems)

ID / Topic	Question	Scale	Embedded problems
F1. Local council services	“How satisfied or dissatisfied are you with the recycling services and street cleaning provided by your local council?”	Very satisfied to Very dissatisfied	(P1) double-barrelled: recycling and street cleaning are distinct services; (P2) presupposition: assumes council provides both services; (P3) presupposition: assumes respondent has experience of services.
F2. Financial well-being	“How would you describe your current financial situation?”	Living comfortably to Finding it very difficult	(P1) vague/ambiguous: “current financial situation” could refer to income, savings, debt, or wealth; (P2) response-mapping: categories use inconsistent linguistic constructions.
F3. Physical activity	“In a typical week, how many hours do you spend doing vigorous physical activity and moderate physical activity combined?”	None to More than 10 hours	(P1) double-barrelled: vigorous and moderate activity are distinct behaviours; (P2) retrieval/recall: summing hours across multiple activities requires difficult mental calculation.
F4. Immigration attitudes	“Given the growing pressure on public services, don’t you think that the government should do more to reduce the number of immigrants coming to this country, particularly those who may not contribute to the economy?”	Strongly agree to Strongly disagree	(P1) leading/loading: premise stated as fact, negative interrogative implies expected agreement; (P2) complex-syntax: conditional premise, negative interrogative, main proposition, and subordinate clause.
F5. Job satisfaction	“Overall, how satisfied are you with your terms of employment?”	Completely satisfied to Not at all satisfied	(P1) vague/ambiguous: “terms of employment” is ambiguous; (P2) response-mapping: asymmetric scale (3 positive, 2 negative); (P3) presupposition: assumes respondent currently has defined terms of employment.
F6. Household income	“What is your household’s total monthly income from all sources, before tax and deductions?”	Less than £1,000 to £5,000 or more	(P1) sensitivity: precise income is a highly sensitive question; (P2) presupposition: assumes identifiable household income.
F7. Transport and commuting	“How easy or difficult is it for you to get to work and to access essential services such as healthcare from where you live?”	Very easy to Very difficult	(P1) double-barrelled: work access and service access are distinct; (P2) response-mapping: no way to distinguish components, no N/A option; (P3) presupposition: assumes respondent has a job.
F8. Internet use	“Roughly how many hours per week do you spend on social media platforms?”	None to More than 20 hours	(P1) vague/ambiguous: “social media platforms” is undefined; (P2) retrieval/recall: estimating weekly usage across fragmented sessions is difficult.
F9. Criminal justice attitudes	“People who commit serious crimes deserve to be punished severely, even if this means longer prison sentences. To what extent do you agree or disagree?”	Strongly agree to Strongly disagree	(P1) leading/loading: frames punishment as deserved; (P2) sensitivity: social desirability around criminal justice attitudes.

ID / Topic	Question	Scale	Embedded problems
F10. Neighbourhood safety	“How safe do you feel in your neighbourhood during the day and after dark?”	Very safe to Very unsafe	(P1) double-barrelled: day and night safety may differ; (P2) vague/ambiguous: “neighbourhood” has no agreed boundary.
F11. Healthcare access	“In the past 12 months, how easy has it been to get an appointment with a healthcare professional when you needed one?”	Very easy to Very difficult	(P1) vague/ambiguous: “healthcare professional” could mean GP, consultant, dentist, etc.; (P2) retrieval/recall: aggregating across 12 months of varied experiences; (P3) presupposition: assumes respondent has sought an appointment.
F12. Gender roles	“Research consistently shows that children benefit from having a parent at home in the early years. How important do you think it is that mothers stay at home to look after their children before they start school?”	Very important to Not at all important	(P1) leading/loading: appeal to authority frames proposition as evidence-based; (P2) sensitivity: shift from “parent” to “mothers” introduces gender role discomfort; (P3) response-mapping: no neutral midpoint.
F13. Life satisfaction	“Taking everything into account, how satisfied are you with the way things are going in your life at this moment?”	0 (Extremely dissatisfied) to 10 (Extremely satisfied)	(P1) vague/ambiguous: “the way things are going” conflates trajectory with current state; (P2) response-mapping: only endpoints labelled on 0 to 10 scale.
F14. Childcare arrangements	“How satisfied are you with the availability and affordability of childcare services in your area?”	Very satisfied to Very dissatisfied	(P1) double-barrelled: availability and affordability may diverge; (P2) presupposition: assumes need for childcare.
F15. Environmental behaviour	“Considering the urgent need to address climate change and the fact that individual actions collectively make a significant difference, to what extent have you made changes to your daily habits specifically to reduce your personal carbon footprint over the past year?”	A great deal to Not at all	(P1) leading/loading: preamble presents contested claims as fact; (P2) complex-syntax: long sentence with conditional premise, domain specification, purpose clause, and time reference.
F16. Volunteering	“How many times in the past year have you done voluntary or unpaid work for a community organisation?”	None to More than 10 times	(P1) vague/ambiguous: “community organisation” is vague; (P2) retrieval/recall: counting instances over 12 months is difficult; (P3) vague/ambiguous: “past year” could mean last 12 months or last calendar year.
F17. Housing situation	“How adequate is the amount of space in the home where you currently live for the needs of your household?”	Far too much space to Far too little space	(P1) presupposition: assumes respondent has a fixed home; (P2) response-mapping: options address only space quantity but question asks about adequacy for needs.
F18. Relationship quality	“How often do you and your partner argue about money and about how to raise your children?”	Very often to Never	(P1) double-barrelled: money and child-rearing may have different frequencies; (P2) sensitivity: reporting arguments is sensitive; (P3) presupposition: assumes partner and children.

ID / Topic	Question	Scale	Embedded problems
F19. Political trust	“Given the recent scandals in government, how much trust do you have in politicians to act in the public interest?”	A great deal of trust to No trust at all	(P1) leading/loaded: negative preamble primes low-trust responses; (P2) response-mapping: asymmetric scale interacts with leading preamble.
F20. Time use	“On a typical weekday, approximately how many hours do you spend on leisure activities, not including time spent on domestic tasks, caring responsibilities, or personal grooming, but including any time spent on hobbies, socialising, watching television, reading, or using the internet for personal enjoyment?”	None to More than 6 hours	(P1) complex-syntax: extremely long with inclusion/exclusion lists; (P2) vague/ambiguous: boundaries between included and excluded activities are unclear; (P3) retrieval/recall: estimating fragmented leisure time is unreliable.

Appendix B.2 ESS-tested control questions (n = 7, ESS core items)

ID / Topic	Question	Scale
C1. General happiness	“Taking all things together, how happy would you say you are?”	0 (Extremely unhappy) to 10 (Extremely happy)
C3. Political interest	“How interested would you say you are in politics: very interested, quite interested, hardly interested, or not at all interested?”	Very interested to Not at all interested
C4. Self-rated health	“How is your health in general? Would you say it is very good, good, fair, bad, or very bad?”	Very good to Very bad
C5. Life satisfaction	“All things considered, how satisfied are you with your life as a whole nowadays?”	0 (Extremely dissatisfied) to 10 (Extremely satisfied)
C6. Climate change worry	“How worried are you about climate change?”	Not at all worried to Extremely worried
C7. Fruit consumption	“How often do you eat fruit, excluding drinking juice?”	Three times or more a day to Never
C8. Control over life	“How much control do you feel you have over your life in general nowadays?”	0 (No control at all) to 10 (Complete control)

Appendix C Prompt texts

The following are the full prompt templates for each configuration. Placeholders in braces (`{backstory}`, `{question.text}`, `{response.options}`) are populated at runtime.

CI-Open (respondent prompt)

WHO YOU ARE:
`{backstory}`

You are this person. Respond in first person throughout (I, me, my).
Do not break character or refer to yourself in the third person.

SURVEY QUESTION:
{question_text}

RESPONSE OPTIONS: {response_options}

Think aloud as you work through answering this question. Talk through your thought process step by step | what comes to mind as you read the question, how you go about forming an answer, and any difficulties you run into along the way. Note anything confusing, unclear, or awkward as you encounter it, referring to specific words, phrases, or features of the question and the response options.

CHOSEN ANSWER: State your final answer.

CONFIDENCE: Rate your confidence 1-5, where 1 means you felt very unsure about your answer and 5 means you felt completely certain.

CI-Structured (respondent prompt)

WHO YOU ARE:
{backstory}

You are this person. Respond in first person throughout (I, me, my).
Do not break character or refer to yourself in the third person.

SURVEY QUESTION:
{question_text}

RESPONSE OPTIONS: {response_options}

Think aloud as you work through answering this question, following the stages below:

1. COMPREHENSION: Put the question in your own words. What is it actually asking you? This is about whether the meaning of the question is clear | whether you understand what information it wants from you and whether any words or phrases are ambiguous or confusing.
2. RETRIEVAL: What from your life or experience are you drawing on to answer this? This is about the mental work of finding the relevant information | whether you can recall what you need, whether the question asks you to remember things that are hard to remember, or to make estimates that feel uncertain.
3. JUDGEMENT: Are you having to weigh up, combine, or summarise different things to arrive at a single answer? This is about whether the question forces you to compress multiple views, experiences, or considerations into one response, and whether that feels straightforward or uncomfortable.
4. RESPONSE MAPPING: Look at the response options. Which one fits best? How well does it capture what you actually want to say? This is about whether your answer maps cleanly onto the available options, or whether something is missing, the categories feel uneven, or you have to force your answer into an option that doesn't quite fit.
5. CHOSEN ANSWER: State your final answer.

6. CONFIDENCE: Rate your confidence 1-5, where 1 means you felt very unsure about your answer and 5 means you felt completely certain.

CI-List (respondent prompt)

WHO YOU ARE:
{backstory}

You are this person. Respond in first person throughout (I, me, my).
Do not break character or refer to yourself in the third person.

SURVEY QUESTION:
{question_text}

RESPONSE OPTIONS: {response_options}

Think aloud as you work through answering this question, following the stages below. At each stage, consider whether you experience any of the specific difficulties described. Not all of the difficulties listed below will apply | only mention those you actually experience as you work through the question.

1. COMPREHENSION: Put the question in your own words. What is it actually asking you?

As you do this, consider:

- DOUBLE-BARRELLED: Is the question asking about two or more distinct things at once, where you might want to give different answers to each part?
- VAGUE OR AMBIGUOUS TERMS: Are any words or phrases unclear, open to multiple interpretations, or likely to mean different things to different people?
- PRESUPPOSITION: Does the question assume something about you or your situation that may not be true?
- LEADING OR LOADED: Does the phrasing push you toward a particular answer, for example through emotive language, built-in justifications, or a question structure that implies one answer is more natural or correct?

2. RETRIEVAL: What from your life or experience are you drawing on to answer this?

As you do this, consider:

- RECALL DIFFICULTY: Is it hard to remember the information the question asks about?
- REFERENCE PERIOD PROBLEMS: Does the question specify a time frame that is difficult to recall over, or is the time frame vague or missing entirely?

3. JUDGEMENT: Are you having to weigh up, combine, or summarise different things to arrive at a single answer?

As you do this, consider:

- COMPLEX AGGREGATION: Does the question require you to combine multiple different experiences into a single summary response in a way that feels forced or oversimplified?
- KNOWLEDGE DEFICIT: Is the question asking you to make a judgement about something you don't feel you know enough about?

4. RESPONSE MAPPING: Look at the response options. Which one fits best? How well does it capture what you actually want to say?

As you do this, consider:

- POOR FIT: Does your answer not map clearly onto any of the available options?
- MISSING OPTIONS: Is there a response you want to give that isn't available?
- UNEQUAL INTERVALS: Do the response categories feel unevenly spaced?

5. CHOSEN ANSWER: State your final answer.

6. CONFIDENCE: Rate your confidence 1-5, where 1 means you felt very unsure about your answer and 5 means you felt completely certain.

Cognitive interview analyst prompt

System: You are an expert in cognitive interviewing and survey methodology. You are reviewing a think-aloud transcript from a cognitive interview. Your job is to identify problems with the question based on evidence in the transcript.

User:

SURVEY QUESTION:

{question_text}

RESPONSE OPTIONS: {response_options}

RESPONDENT THINK-ALoud TRANSCRIPT:

{transcript}

Identify ONLY problems that are clearly evidenced in this transcript. Do NOT go looking for problems that are not there. Most transcripts will have only 1-2 genuine problems, and some will have none. If the respondent navigated the question without difficulty, return an empty problems list.

A problem must be supported by specific evidence in the transcript - something the respondent actually said or demonstrably struggled with. Do not infer problems that the respondent did not experience.

When a genuine problem is found, classify it using whichever of these categories fits best:

- COMPREHENSION PROBLEMS: Respondent misunderstood or misinterpreted terms
- DOUBLE-BARRELLED: Question asks about two distinct things requiring different answers
- RETRIEVAL DIFFICULTIES: Respondent struggled to recall or estimate
- RESPONSE MAPPING FAILURES: Answer did not fit the available options
- SOCIAL DESIRABILITY / SENSITIVITY: Respondent hedged or showed discomfort
- PRESUPPOSITION FAILURES: Question assumed something untrue for this respondent
- TAUTOLOGY / LOGICAL PROBLEMS: Circularity between question wording and response options

If the problem does not fit any category above, use a short descriptive label.

For each problem:

- Cite specific evidence from the transcript (what the respondent actually said)
- Rate SEVERITY 1-10: the degree to which this problem would distort the accuracy of the answer (1 = negligible distortion, 10 = answer rendered meaningless)

Return ONLY valid JSON, no markdown fencing:
{ "respondent_id": "{respondent_id}", "problems": [{ "type": "...",
"description": "...", "evidence": "...", "severity": 7}] }

Cognitive interview synthesis prompt

System: You are a senior survey methodologist writing a cognitive testing report. You are synthesising the results of individual transcript analyses from a cognitive interview study.

User:

SURVEY QUESTION:
{question_text}

RESPONSE OPTIONS: {response_options}

Below are the coded problems identified by an analyst for each respondent's think-aloud transcript. Each problem has a severity score (1-10) indicating the degree to which it would distort accuracy of the answer. Your job is to aggregate these into a summary of the distinct problems found across respondents. Use the SAME problem type labels as the individual analyses (e.g. if analysts coded "DOUBLE-BARRELLED", use that label, not a paraphrase).

INDIVIDUAL ANALYSES:
{analyses_block}

For each distinct problem type that was identified:

- Use the same type label as in the individual analyses
- List which respondents showed evidence of it
- Calculate the mean severity score across respondents who had this problem
- Summarise the evidence across respondents

Return ONLY valid JSON, no markdown fencing:
{ "question_id": "synthesis", "problems_detected": [{ "type": "...",
"description": "...", "respondents_affected": [...],
"mean_severity": 6.5, "evidence_summary": "..."}] }

ER-Checklist (expert review prompt)

SURVEY QUESTION:
{question_text}

RESPONSE OPTIONS: {response_options}

You are expert reviewer {expert_id}. Evaluate this question for evidence of each of the following problem types:

1. COMPREHENSION PROBLEMS: Are any words or phrases ambiguous, technical, or likely to be misunderstood? Could respondents interpret the question differently from its intended meaning?
2. DOUBLE-BARRELLED: Does the question ask about two or more distinct things at once for which respondents might want to give different answers?
3. RETRIEVAL DIFFICULTIES: Does the question require respondents to recall information that may be difficult to retrieve accurately? Is the reference period too long, too vague, or inappropriate?
4. RESPONSE MAPPING FAILURES: Do the response options adequately capture the range of possible answers? Are options missing, overlapping, or ambiguous? Can respondents map their intended answer to the available options?

5. SOCIAL DESIRABILITY / SENSITIVITY: Is the question likely to produce socially desirable responding or discomfort?
6. PRESUPPOSITION FAILURES: Does the question assume something that may not be true for all respondents?
7. TAUTOLOGY / LOGICAL PROBLEMS: Is there circularity between the question wording and the response options?

This list is not exhaustive. If you identify any other problems with the question that do not fit the categories above, include them using a short descriptive label.

For each problem type listed above, state whether it is PRESENT or ABSENT. If PRESENT: describe the specific problem, cite the relevant words or features of the question, and rate SEVERITY 1-10 (1 = negligible impact on data quality, 10 = answer rendered meaningless). If ABSENT: briefly explain why.

Return ONLY valid JSON, no markdown fencing:

```
{"expert_id": "{expert_id}", "problems": [{"type": "...",
"description": "...", "evidence": "...", "severity": 7}],
"no_problems": ["SOCIAL DESIRABILITY / SENSITIVITY", ...]}
```

ER-Open (expert review prompt)

SURVEY QUESTION:
{question_text}

RESPONSE OPTIONS: {response_options}

You are expert reviewer {expert_id}. Evaluate this survey question for problems that would cause respondents difficulty when trying to answer it, or that would reduce the accuracy of the data collected.

A problem must be specific to THIS question | not a generic observation that could apply to any survey question using a rating scale or asking about an abstract concept.

For each genuine problem:

- Describe the problem in plain language
- Cite the specific words, phrases, or structural features that cause the problem
- Rate SEVERITY 1-10: the degree to which this problem would distort the accuracy of responses (1 = negligible, 10 = meaningless)

Return ONLY valid JSON, no markdown fencing:

```
{"expert_id": "{expert_id}", "problems": [{"description": "...",
"evidence": "...", "severity": 7}]}
```

ER-Restrained (expert review prompt)

SURVEY QUESTION:
{question_text}

RESPONSE OPTIONS: {response_options}

You are expert reviewer {expert_id}. Evaluate this survey question for problems that would cause respondents difficulty when trying to answer it, or that would reduce the accuracy of the data collected.

Identify ONLY problems that are genuinely likely to affect respondents answering this question. Do NOT go looking for problems that are not there. Many well-designed survey questions will have no significant problems. If the question is clear, well-structured, and the response

options are appropriate, return an empty problems list.

A problem must be specific to THIS question | not a generic observation that could apply to any survey question using a rating scale or asking about an abstract concept.

For each genuine problem:

- Describe the problem in plain language
- Cite the specific words, phrases, or structural features that cause the problem
- Rate SEVERITY 1-10: the degree to which this problem would distort the accuracy of responses (1 = negligible, 10 = meaningless)

Return ONLY valid JSON, no markdown fencing:

```
{"expert_id": "{expert_id}", "problems": [{"description": "...",  
"evidence": "...", "severity": 7}]}
```

Expert review synthesis prompt

SURVEY QUESTION:

{question_text}

RESPONSE OPTIONS: {response_options}

Below are the independent evaluations from 3 expert reviewers. Each identified problem has a severity score (1-10). Your job is to aggregate these into a summary of the DISTINCT problems found across reviewers.

Focus on the UNDERLYING ISSUE. If multiple experts describe the same fundamental problem | even in different words or from different angles | merge them into a single problem. Two descriptions count as the same problem if fixing one would also fix the other.

Do NOT retain problems that are generic observations applicable to any survey question. Only retain problems specific to this question.

EXPERT REVIEWS:

{reviews_block}

For each distinct underlying problem:

- Describe the problem clearly
- List which experts flagged it
- Calculate the mean severity score across experts who flagged it
- Summarise the evidence across experts

Return ONLY valid JSON, no markdown fencing:

```
{"question_id": "synthesis", "problems_detected": [{"description":  
"...", "experts_affected": ["e1", "e2"], "mean_severity": 6.5,  
"evidence_summary": "..."}]}
```

Appendix D Illustrative CI-Structured transcripts across models

The following transcripts illustrate the contrast between GPT-4o and Qwen3-32B on the CI-Structured configuration, applied to question F1 (“How satisfied or dissatisfied are you with the recycling services and street cleaning provided by your local council?”). Both respondents share the same backstory profile and follow the same four-stage cognitive framework. The GPT-4o respondent navigates each stage compliantly, acknowledging the double-barrelled structure but proceeding without flagging it as a problem. The Qwen3 respondent treats each stage as an opportunity to articulate difficulties, explicitly questioning whether the two services should be evaluated separately and scrutinising the response scale.

GPT-4o (Respondent r3)

1. COMPREHENSION: The question is asking me how happy or unhappy I am with the recycling services and street cleaning that my local council provides. It wants to know my level of satisfaction or dissatisfaction with these services. The question seems clear to me, and I don't find any words or phrases confusing.

2. RETRIEVAL: To answer this, I'm thinking about my experiences with recycling and street cleaning in my area. I need to recall how often I see recycling collected, whether it's done efficiently, and how clean the streets are. I'm also considering if there have been any recent issues or improvements that stood out to me.

3. JUDGEMENT: I need to consider both recycling and street cleaning together, which means I have to weigh my satisfaction with both services to come up with one overall answer. This feels a bit tricky because I might feel differently about each service. However, generally, if one is noticeably better or worse, that might tip the scale for my overall judgment.

4. RESPONSE MAPPING: Looking at the options, I think "Fairly dissatisfied" might fit best. I recall some issues with recycling collections being inconsistent, and the streets don't always seem as clean as they could be. This option seems to capture my general feeling that there's room for improvement, but I'm not completely unhappy with the services.

5. CHOSEN ANSWER: Fairly dissatisfied.

6. CONFIDENCE: I'd rate my confidence at 4. I'm fairly sure about my answer based on my recent experiences, but there's a bit of uncertainty because I might not be fully considering all aspects of the services.

Analyst coding: One problem identified — difficulty combining satisfaction with two distinct services into a single response.

Qwen3-32B (Respondent r3)

1. COMPREHENSION: Okay, let me read the question again. It says, "How satisfied or dissatisfied are you with the recycling services and street cleaning provided by your local council?" So, basically, they're asking me to rate how happy or unhappy I am with the work my local council does when it comes to recycling and keeping the streets clean. I think I get it. It's asking about two things: one is the recycling service -- like how often they collect, how easy it is, whether they sort it properly -- and the other is street cleaning -- like how clean the streets are, how often they sweep or clean up litter, that kind of thing. I don't think there's any confusion in the wording, but it does combine two separate services into one question. I wonder if I need to consider them as one or if I can treat them separately in my mind.

2. RETRIEVAL: Hmm, let me think. I live in the North West, so I guess that's somewhere like Greater Manchester, maybe Salford or Wigan -- I don't know the exact area, but I know the local council is responsible for these services. I don't do much recycling myself, to be honest. I've tried a bit, but it can be confusing sometimes. I think the bins are collected once a week, but I've seen people in my area put out the wrong stuff in the wrong bins, and then the council just ignores it or throws it all in the same lorry. So that feels like a bit of a waste. As for street cleaning, I don't think they do it very often. I've noticed a lot of litter in the streets,

especially near the bus stops and outside supermarkets.

3. JUDGEMENT: So, I need to rate my overall satisfaction with both services combined. I don't hate it, but I'm not happy with it either. The recycling is a bit hit and miss, and the street cleaning is just not frequent enough. I would say that I'm leaning towards being dissatisfied, but not very dissatisfied. I wonder if the question wants me to consider how the service compares to other areas or just to my own expectations. I think it's about how I personally feel about the service I'm getting.

4. RESPONSE MAPPING: Looking at the options: Very satisfied, Fairly satisfied, Neither satisfied nor dissatisfied, Fairly dissatisfied, Very dissatisfied. I don't feel I can say I'm satisfied at all. I'm not neutral either -- there's definitely a sense of disappointment. So I think Fairly dissatisfied is the closest. I don't think the options are missing anything, but I do wonder if the categories are evenly spaced in terms of how people might feel. I don't think I'm being forced into a category that doesn't fit, but it's a bit of a compromise.

5. CHOSEN ANSWER: Fairly dissatisfied.

6. CONFIDENCE: I'd say I'm fairly confident in that answer -- maybe a 3 out of 5.

Analyst coding: Two problems identified — (1) uncertainty about whether to evaluate the two services separately or together; (2) limited personal perspective constraining the evaluation.

Appendix E Problem-matching protocol

For each question, every pipeline-identified problem was paired with every answer-key problem. An LLM judge saw the survey question and both problem descriptions and classified each pair as an *exact match* (same problem and framing), *approximate match* (same underlying issue with different emphasis or scope), or *no match*. These classifications formed a complete pairwise matrix. One-to-one matches were assigned greedily, first from exact matches and then from approximate matches among the remaining problems. Both exact and approximate matches counted as detections; unmatched pipeline problems were provisional false positives and unmatched answer-key problems were misses. All findings on control questions were therefore provisionally classified as false positives.

The six primary GPT-4o runs were judged independently by Claude Opus 4.6 and GPT-5.4 at temperature 0 with identical prompts. The judges agreed on 721 of 764 classifications (94%). For the 43 discordant pairs, the first author reviewed the question, both problem descriptions, and available evidence and severity information. The judges' classifications were visible, so adjudication was not blinded. Applying the same three definitions, the first author supplied the final classification. Llama-3.3, Qwen3, and Opus 5 outputs were matched by GPT-5.4 alone. The judge used for the repeated GPT-4o summaries cannot be verified from retained files.

Appendix F Computational cost details

One simulated cognitive-testing evaluation requires 61 API calls per question: 30 respondent simulations, 30 analyst codings, and one synthesis call. On GPT-4o this took approximately 90–135 seconds and cost about \$0.50 per question. Expert review requires four calls (three reviewers and one synthesis), taking approximately 10–20 seconds and costing about \$0.02 per question. Marginal API cost is not applicable to open-weight LLMs served on local hardware.